

HOUSE PRICE PREDICTION ANALYSIS REPORT

Dataset Used: <https://www.kaggle.com/datasets/yasserh/housing-prices-dataset>

Objective

The objective of this project was to build a machine learning model to predict house prices based on various property features and identify the factors that have the greatest impact on pricing.

Dataset Overview

The dataset contains information about residential properties, including area, number of bedrooms, bathrooms, parking spaces, air conditioning, furnishing status, and other housing-related features. The target variable for prediction was the house price.

Data Preprocessing

The dataset was checked for missing values and duplicate records. Categorical variables were converted into numerical form using encoding techniques to make them suitable for machine learning models.

Model Building and Evaluation

Two machine learning models were trained and evaluated:

- Linear Regression
- Random Forest Regressor

The models were compared using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score. Visualizations such as a price distribution histogram, correlation heatmap, and scatter plot were also done to analyse the data and model performance.

Results

Model	MAE	RMSE	R^2 Score
Linear Regression	970,043	1,324,507	0.653
Random Forest	1,021,546	1,400,566	0.612

Key Insights

The analysis showed that house area, number of bathrooms, and air conditioning were the most important factors affecting house prices. Among the two models tested, Linear Regression performed better explaining about 65% of the variation in house prices. The model's predictions differed from actual prices by approximately ₹9.7 lakh on average. An interesting finding was that Linear Regression outperformed Random Forest, suggesting that the relationship between house features and prices is fairly linear.

Based on these results, real estate businesses should focus on larger properties and key amenities such as bathrooms and air conditioning when practising and marketing houses.

Conclusion

The project successfully developed a house price prediction model and identified the most influential factors affecting property values. The findings can help support pricing decisions and provide useful insights for real estate businesses.